



INF1006 - Assignment

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1. Introduction

This project focuses on designing and simulating an enterprise network that connects the E2 and W3 buildings at the SIT campus using Cisco Packet Tracer. The goal is to ensure reliable communication between the labs, provide access to internal services such as DNS and the web server, and enable Internet connectivity through an ISP using the assigned network 103.31.24.0/23.

The network is designed using a hierarchical structure. E2 houses SSNL, CSL, and the central server room, while W3 includes W3 Lab 1, W3 Lab 2, and the POD office. Routers manage communication between buildings and subnets, while switches connect devices within each lab. VLANs are used to separate departments into different broadcast domains for better organization and security, and wireless access points are deployed as required.

2. Network Design Overview

Core Layer

Each building deploys one main router and one backup router. The E2 main router connects to the ISP, while backup routers provide failover to eliminate single points of failure. Redundant inter-building links maintain connectivity during outages.

Server Infrastructure

Two servers are located in E2: DNS Server (domain resolution), Web Server (www.singaporetech.edu.sg), Backup Server (redundancy for DNS and Web services)

Distribution Layer

Each building contains one main and one backup distribution switch. Access switches connect redundantly to both. STP is enabled to prevent loops and activate backup paths when required.

Access Layer

Each lab has two main switches. The POD office has one main. This ensures switch-level redundancy.

3. VLAN Segmentation Strategy

To reduce broadcast domains and enhance security, VLAN segmentation was implemented across both E2 and W3 buildings. Each department operates within its own VLAN (Figure 2). This segmentation confines broadcast traffic within departments and allows inter-VLAN communication to be controlled through Layer 3 routing and access policies. The assigned network block 103.31.24.0/23 was subnetted using VLSM to allocate address space efficiently according to departmental host requirements (Figure 2). Larger departments were assigned /26 subnets, medium-sized labs /27, and smaller management networks smaller subnets. A /30 subnet was used for the router-to-ISP link to conserve address space. This design minimizes IP wastage, supports scalability, and ensures efficient traffic isolation across the enterprise network.

4. Centralised Services Design

A dedicated Server VLAN was implemented to host the DNS and Web servers, allowing users across the network to access services using domain names such as www.singaporetech.edu.sg and www.networkassignment.com. Centralising these services improves overall manageability by keeping critical infrastructure in a single controlled environment. Isolating the servers in their

own VLAN also enhances security and enables controlled access from other VLANs through routing and policy enforcement.

5. Connectivity Verification

A series of validation tests were conducted to ensure the network met all functional requirements. These included inter-VLAN ping tests, inter-building communication verification, external ISP reachability checks, DNS resolution testing, and web access validation using domain names. All tests were successful, confirming that the network operates correctly and satisfies the specified connectivity requirements.

6. Static Routing

Static routing was selected over dynamic routing protocols such as OSPF or RIP because the network topology is relatively small, stable, and predictable. Since routes are unlikely to change frequently, static routing provides a simple and deterministic solution without the additional overhead associated with routing protocol updates and recalculations. It also simplifies troubleshooting, as traffic paths are manually defined and easily traceable. Furthermore, dynamic routing was not required by the assignment specification, making static routing a more appropriate and controlled choice for this design.

7. System Design Justification

The network design is designed with having the lowest cost and reducing the chance of any system failure in mind. Redundancy is implemented at multiple layers through dual routers, backup distribution switch and a backup server. This multi-layer approach eliminates a single point of failure and ensures service continuity for critical resources such as DNS and web service.

VLAN segmentation is used to isolate labs, offices, and servers into separate broadcast domains, improving security and reducing unnecessary broadcast traffic. Inter-VLAN routing at the core layer enables controlled communication within each building while supporting policy enforcement, such as printer access restrictions.

This hierarchical design also supports scalability and efficient IP utilization through VLSM, allowing future expansion without major redesign. Overall, the network reflects practical enterprise design principles while maintaining a reasonable balance between hardware deployment and reliability.

DHCP services were deployed on the central server in E2 instead of the router to follow enterprise design principles. This ensures role separation, improves scalability, allows centralised management of IP addressing across multiple VLANs, and supports future expansion. Router resources are preserved for routing and WAN connectivity, improving overall network performance and maintainability.

APPENDIX

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
ISP	G0/0	Public IP	Public Mask	NA
	G0/2	200.1.1.1	255.255.255.0	NA
	S0/1/0	10.0.0.2	255.255.255.252	NA
	S0/1/1	10.0.0.6	255.255.255.252	NA
E2 Router	G0/0.10	103.31.24.66	255.255.255.192	103.31.24.65 (HSRP)
	G0/0.20	103.31.24.2	255.255.255.192	103.31.24.1 (HSRP)
	G0/0.60	103.31.24.194	255.255.255.224	103.31.24.193 (HSRP)
	S0/0/0	172.16.0.1	255.255.255.252	NA
	S0/0/1	10.10.20.1	255.255.255.252	NA
	S0/1/0	10.0.0.1	255.255.255.252	NA
E2 Backup Router	G0/0.10	103.31.24.67	255.255.255.192	103.31.24.65(HSRP)
	G0/0.20	103.31.24.3	255.255.255.192	103.31.24.1(HSRP)
	G0/0.30	103.31.24.131	255.255.255.224	103.31.24.129(HSRP)
	G0/0.40	103.31.24.163	255.255.255.224	103.31.24.161(HSRP)
	G0/0.50	103.31.24.227	255.255.255.248	103.31.24.225(HSRP)
	G0/0.60	103.31.24.195	255.255.255.224	103.31.24.193(HSRP)
	S0/0/0	172.16.0.5	255.255.255.252	NA
	S0/0/1	10.10.20.2	255.255.255.252	NA
	S0/3/1	10.0.0.5	255.255.255.252	NA
W3 Router	G0/0.30	103.31.24.130	255.255.255.224	103.31.24.129(HSRP)
	G0/0.40	103.31.24.162	255.255.255.224	103.31.24.161(HSRP)
	G0/0.50	103.31.24.226	255.255.255.240	103.31.24.225(HSRP)

	S0/0/0	172.16.0.2	255.255.255.252	NA
	S0/0/1	10.10.10.1	255.255.255.252	NA
W3 Backup Router	G0/0.30	103.31.24.131	255.255.255.224	103.31.24.129(HSRP)
	G0/0.40	103.31.24.163	255.255.255.224	103.31.24.161(HSRP)
	G0/0.50	103.31.24.227	255.255.255.240	103.31.24.225(HSRP)
DNS Server	FastEthernet0	103.21.24.202	255.255.255.224	103.31.24.193
Web Server	FastEthernet0	103.31.24.200	255.255.255.224	103.31.24.193
Internet Web Server	FastEthernet0	200.1.1.10	255.255.255.0	200.1.1.1
SIT website		103.31.24.200		
Network website		200.1.1.10		

Figure 1: Addressing Table

103.31.24.64/26 - VLAN 10	SSNL
103.31.24.0/26 - VLAN 20	CSL
103.31.24.128/27 - VLAN 30	W3 Lab 1
103.31.24.160/27 - VLAN 40	W3 Lab 2
103.31.24.224/29 - VLAN 50	POD Management
103.31.24.192/27 - VLAN 60	Server Required for DNS/Web

Figure 2: IP Allocation for VLAN departments